CONSUMER ATTENTION MAPPING SYSTEM (CAMS)

AI Video Analytics & Crowd Intelligence Report

Video File: video_1785612530551_969535342.mp4 | **Generated:** 2026-08-01 19:28:56

1. Executive Summary

Automated AI vision analysis using YOLOv8 was completed for **video_1785612530551_969535342.mp4**. The video was processed across **120 frames** (Duration: 0m 4s). During the analysis, **28 unique shoppers** were tracked with a peak crowd density of **16 individuals**. Overall crowd density was evaluated at **Critical level (100.0%)** with an average detection confidence of **89.4%**.

2. Key Analytics Summary Table

Metric Parameter	Recorded Value	Benchmark / Context
Total Detection Instances	48	Cumulative person bounding boxes
Unique Tracked People	28	Individual shopper trajectories
Maximum Crowd (Peak)	16	Highest simultaneous count in single frame
Average Crowd Density	0.4 people/frame	Mean spatial occupancy
Crowd Occupancy Rate	100.0%	Density Level: Critical
Average AI Confidence	89.4%	YOLOv8 model detection accuracy
Total Frames Processed	120	Video Duration: 0m 4s
YOLOv8 Processing Time	1.27 seconds	Inference execution duration

3. Crowd & Spatial Statistics

The spatial crowd level is rated **Critical**. Average simultaneous occupancy stood at 0.4 shoppers per frame. Threshold analysis indicates that floor circulation was within acceptable limits without major bottleneck congestion.

4. Object Detection Breakdown

Object Category	Total Detections
Person / Shopper	48
Shopping Cart / Basket	9
Other Objects / Items	24

5. AI Recommendations & Strategic Actions

• Optimize shelf display placement near high-traffic bottleneck zones to increase dwell engagement.

- Crowd level is currently Critical (100.0%). Ensure staff allocation aligns with peak traffic.
- Consider rebalancing aisle width near high dwell areas to smooth customer movement during busy windows.
- Maintain optimal camera angle (1080p, 30FPS) for maximum YOLOv8 detection accuracy.